# First Best Implementation with Costly Information Acquisition*

Daniil Larionov† Hien Pham‡ Takuro Yamashita§

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**Abstract**

We study a mechanism design model with flexible but costly information acquisition. There is a principal and $I \geq 4$ agents. The principal and the agents share a common prior over the set of payoff-relevant states of the world. The principal proposes a mechanism to the agents who can then acquire information about the state of the world by privately designing a signal device. As long as it is costless for each agent to acquire a signal that is pairwise independent from the state of the world, we show that there exists a mechanism which allows the principal to implement any social choice rule at zero information acquisition cost to the agents.

## 1 Introduction

In most mechanism design problems, there is a collection of agents who have exogenously given private information, and there is a principal who desires to implement a social choice rule,

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†Department of Economics, University of Mannheim. daniil.larionov@gess.uni-mannheim.de

‡Toulouse School of Economics, University of Toulouse Capitole. hien.pham@tse-fr.eu

§Toulouse School of Economics, University of Toulouse Capitole. takuro.yamashita@tse-fr.eu

by designing a mechanism which incentivizes the agents to reveal their information.

In many problems in practice, however, the agents' private information is often a consequence of their own (possibly costly) *information acquisition*. For example, bidders in an oil-tract auction (Wilson, 1969) may conduct test drills; bidders in a spectrum auction may conduct market research; voters in a presidential election may investigate the candidates' past political activities.

Importantly, in such situations, a mechanism in place does not only affect each agent's incentive of reporting the acquired information truthfully, but also affects his choice of *what kind of information* to acquire. In this sense, the properties of desirable mechanisms could potentially be very different from those which only guarantee truth-telling incentives for a given information structure.

Although this issue is already relevant in single-agent environments,[1] the degree of complexity is even higher in a multi-agent environment: in principle, flexibility of each agent's information acquisition action does not only mean flexibility in terms of his signal's informativeness about the payoff-relevant state (e.g., the amount of oil in a tract), but also means flexibility in terms of his opponents' signals. This issue of higher-order information/beliefs is unique in multi-agent environments.[2] As far as we understand, when it comes to higher-order information, there is no obvious or universally agreed upon class of information acquisition cost functions [3] Hence, in this paper, we take an agnostic approach by essentially assuming that an agent's information cost only depends on its informativeness about the payoff-relevant state, but not on the others' signals. We believe that it is a reasonable "first-step" assumption, but we must admit that our result crucially depends on this assumption. We discuss it in Section 5.1.

We consider a model with four or more agents, and assume that it is costless for each agent to acquire a signal that is pairwise independent from the state of the world.[4] The principal and

[1]Mensch (2020) studies a mechanism design problem with a single agent. See also Section 1.1

[2]Of course, this issue can be avoided if we impose some *ad hoc* assumptions such as conditional independence, although that is not the scope of the paper.

[3]See Section 1.1 and 5.1.

[4]The literature about cost of information proposes and discusses a variety of possible cost functions (see Section 1.1), but it seems to be a usual property that uninformative signals about the state of the world are

agents share a common prior, and none of them has any private information at the beginning. We show that there exists a mechanism which allows the principal to implement any social choice rule at zero information acquisition cost to the agents. The mechanism recommends each agent to choose a special information acquisition action, which satisfies the *individually-uninformative-but-aggregately-revealing* property of Zhu (2021) (and each agent finds it optimal to obey this recommended action). The *individually-uninformative* part means that each agent's signal on its own is pairwise independent from the payoff-relevant state, which guarantees that his information cost is zero. The *aggregately-revealing* part means that the principal, by observing all the agents' reports — in fact, any two of them, — can correctly identify the true payoff-relevant state. The fact that only two is enough, together with the fact that there are four or more agents, enables the principal to detect any unilateral deviation. It establishes the incentive compatibility of the mechanism.

## 1.1 Related Literature

In the literature of information acquisition in mechanism design, we usually consider restricted/less flexible spaces of information (see, for example, Bergemann and Välimäki (2002), Shi (2012) and Bikhchandani and Obara (2017)).

Mensch (2020) studies mechanism design with a single agent's flexible and costly information acquisition, building on the rational inattention framework (Sims (2003)). Flexible information acquisition is also considered by Roesler and Szentes (2017) in the context of buyer-optimal information in monopoly pricing,[5] by Bergemann, Brooks, and Morris (2017) and Brooks and Du (2021) in the context of seller-pessimal information in common-value auction, and by Yamashita (2018) in private-value auction. All these papers have a common feature that there is a single entity, "nature", who chooses the information structure (of one or multiple agents). In contrast, in our model, each of multiple agents acquires information in a decentralized manner, leading to a very different conclusion.

costless.

[5]See also Condorelli and Szentes (2020), though they also consider non-information changes of the agent's private information distribution.

The information structure we employ was proposed in the context of mechanism design by Zhu (2021), who studies information disclosure by a mechanism designer. It builds on the idea of the one-time pad, unbreakable encryption (Shannon, 1949).[6]

This key information structure makes the agents' acquired information statistically dependent. In quasi-linear environments, Crémer and McLean (1988) show that the principal can extract full surplus from the agents. Although the extreme positivity of the results is a common feature of our paper and theirs, the two problems are quite different. First, our paper does not assume quasi-linearity. Second, their side-bet mechanism exploits an exogenously given correlated signal structure, and it is not clear if it can be induced as an equilibrium information structure given some reasonable spaces of information acquisition actions.[7] In our case, the above mentioned information structure is indeed an equilibrium outcome, even though each agent can potentially acquire information independent from the others' signals.

There is a growing literature about cost of flexible information in decision environments. See for example Sims (2003), Matejka and Mckay (2015), Caplin and Dean (2015), and Pomatto, Strack, and Tamuz (2020). Usually the main focus is on the cost of acquiring more/less precise information about the payoff-relevant state, and its relationship with a single decision maker's optimal choice. The framework, however, has been applied in multi-player problem (i.e., in games), such as in coordination games (Yang (2015); Morris and Yang (2021); Denti (2020). In particular, Denti (2020) proposes a model of unrestricted information acquisition in games in which, as in our paper, the players can endogenously learn about a payoff-relevant state and actions of other players.

[6]See also Kalai, Kalai, Lehrer, and Samet (2010); Renou and Tomala (2012); Renault, Renou, , and Tomala (2014), in the context of games of communication network. Peters and Troncoso-Valverde (2013) apply this idea in mechanism-design games with multiple principals, and Liu (2015) applies it in his concept of individually uninformative correlating device. Our construction is most directly related to Zhu (2021).

[7]Bikhchandani (2010) shows that, indeed, an agent in the Crémer-McLean mechanism may have a strong incentive of acquiring information about others.

# 2 Model

## 2.1 Setup

There exist a principal and $I \geq 4$ agents, and a finite set of payoff-relevant states $\Theta$. Each agent $i$'s payoff is denoted $u_i(d, \theta)$, when a social decision $d \in D$ is selected in state $\theta$.[8] For example, in an auction, $d$ may comprise a vector of each bidder $i$'s winning probability and his expected payment, and $u_i$ is quasi-linear in the payment part. Later, his payoff *net* his information acquisition cost is considered as his objective.

At the beginning, neither the principal nor any of the agents know $\theta$, but they share a common prior $\mu_0$ on $\Theta$. In what follows, it is useful to imagine that there exists an unobservable fundamental state of nature, $x \in X$, where $X$ is a measurable space associated with a probability measure $\mathbb{P}$. We assume $X = [0, 1]$ with a Borel $\sigma$-algebra and $x$ is uniformly distributed on $[0, 1]$.[9] Taking a richer space of the fundamental state of nature does not change our results. Then, we define $\theta$ as a measurable function of $x$, $\theta = \theta(x)$. $\theta \sim \mu_0$ means that $\int_0^1 1_{\{\theta(x)=\theta\}} dx = \mu_0(\theta)$ for each $\theta \in \Theta$.

The agents can acquire *costly* information about $\theta$ by generating private and possibly correlated signals, whereas the principal cannot. More specifically, let $S_i$ be the set of possible signals agent $i$ may potentially obtain by any information acquisition technology. Potentially, $S_i$ (in particular, its size) may be a part of $i$'s choice, but assuming exogenous $S_i$ simplifies our analysis. Besides, as long as $|S_i| \geq |\Theta|$, the same results obtain. $i$'s information acquisition action is a measurable function $\sigma_i : [0, 1] \to S_i$, such that, once $x$ (and hence $\theta = \theta(x)$) is realized, then $i$ observes $s_i = \sigma_i(x)$. Let $\Sigma_i$ denote the set of all such measurable functions, defining $i$'s information acquisition action space. We assume that information acquisition is fully private in the sense that neither the principal nor any other agent observes which action $i$ takes and which signal is realized. Agent $i$'s objective is the *net* payoff $u_i(d, \theta) - c_i(\sigma_i)$. We assume the following property of the cost function:

[8] We can endow the principal with his own payoff function $u_0(q, \theta)$, though it is not necessary.

[9] See Gentzkow and Kamenica (2017) who adapt this approach in the context of multi-sender Bayesian persuasion.

**Assumption 1.** *$c_i(\sigma_i) \geq 0$ for any $\sigma_i \in \Sigma_i$. $c_i(\sigma_i) = 0$ if its induced $s_i$ and $\theta$ are independently distributed, that is, for each $s_i \in S_i$ and $\theta \in \Theta$:*

$$\int_0^1 1_{\{\sigma_i(x)=s_i \text{ and } \theta(x)=\theta\}} dx = \left(\int_0^1 1_{\{\sigma_i(x)=s_i\}} dx\right)\left(\int_0^1 1_{\{\theta(x)=\theta\}} dx\right)$$

It is a quite general specification, which includes many popular cost functions as special cases: entropy-based cost functions, more general posterior-separable cost functions, and even some posterior-non-separable ones.[10] However, it should be noted that we do not allow $i$'s information acquisition cost to depend on others' information acquisition actions. In other words, the cost of information acquisition for each agent depends only on how much this agent individually learns about $\theta$, and not on how much other agents learn. We think it is a reasonable choice, especially as a first-step approach, but it is a restriction, and our result hinges on it.[11]

Given any $\sigma$, a joint distribution for $(\theta, s)$ is determined, which we denote by $\alpha \in \Delta(\Theta \times S)$. If we want to make its dependence on $\sigma$ more explicit, we denote it as $\alpha_\sigma$.

## 2.2 Mechanism

The principal faces both hidden action and hidden information of each agent. The principal can commit to a mechanism at the ex ante stage in order to control the agents' incentives. More specifically, following the literature, we let the principal (i) send a message privately to each agent before his information acquisition action, and (ii) collect a message privately from each agent after his observing a signal. Formally, a mechanism comprises $(R, \rho; M, \delta)$ where $R = (R_i)_{i=1}^I$ and $M = (M_i)_{i=1}^I$; $R_i$ denotes the set of messages that the principal can send to each agent $i$; $M_i$ denotes the set of messages that each agent $i$ can send to the principal; $\rho \in \Delta(R)$ is a distribution over the principal's messages, and $\delta : R \times M \to D$ denotes the decision rule.

[10] See the literature of cost of information, such as Sims (2003), Matejka and Mckay (2015), Caplin and Dean (2015), and Pomatto, Strack, and Tamuz (2020).

[11] For example, Strulovici (2021) considers a situation where some hard information is scarce; Denti (2020) considers a general interdependent information cost. As far as we know, there is no general consensus in the literature which types of correlation structure should be more or less costly than others. Perhaps it depends on specific contexts of interest. See also our discussion in the concluding section.

The timing of the game is summarized as follows:

$t = 0$: $x \sim U(0,1)$ is drawn (though no one observes it).

$t = 1$: The principal designs a mechanism $(R, \rho; M, \delta)$.

$t = 2$: After observing the mechanism and receiving $r_i \in R_i$, each agent $i$ privately chooses his information acquisition action $\sigma_i \in \Sigma_i$.

$t = 3$: Each agent $i$ privately observes $s_i = \sigma_i(x)$, and privately sends $m_i \in M_i$ to the principal.

$t = 4$: The principal executes $d = \delta(r, m)$ where $m = (m_i)_{i=1}^I$.

Because no agent observes the other agents' actions or information (even noisily) at all, we consider Bayesian equilibrium as a solution concept. Then, applying the revelation principle of Forges (1986), we focus on *direct* mechanisms where (i) the principal directly recommends an information-acquisition action to each agent, and each agent directly reports a signal to the principal, and (ii) each agent finds it optimal to obey the recommended action and truthfully report his signal.[12]

Formally, a direct mechanism comprises $((\sigma_i)_{i=1}^I, \delta)$, where the principal recommends $\sigma_i \in \Sigma_i$ privately to each agent $i$,[13] and executes $\delta(s) \in D$ if the agents report $s = (s_i)_{i=1}^I \in S$. It is a feasible direct mechanism if it satisfies the following constraints: for any $\sigma_i' \in \Sigma_i$ and $\tau_i : S_i \to S_i$,

$$\sum_{\theta, s_i, s_{-i}} (u_i(\delta(s_i, s_{-i}), \theta)\alpha_{\sigma_i, \sigma_{-i}}(\theta, s_i, s_{-i})) - c_i(\sigma_i) \geq \sum_{\theta, s_i, s_{-i}} (u_i(\delta(\tau_i(s_i), s_{-i}), \theta)\alpha_{\sigma_i', \sigma_{-i}}(\theta, s_i, s_{-i})) - c_i(\sigma_i').$$

[12]The proof proceeds as follows. First, imagine an auxiliary game where there is no principal, but instead, there is a fictitious player ("player 0") who is indifferent across all decisions in any state. At first, each agent $i$ plays $\sigma_i$ privately, and then observes the realized signal $s_i$ privately. Then, (without any communication), player 0 chooses $d \in D$. Interpreting this as a baseline extensive-form game, it is easy to see that our current game (with the principal) is the mediated communication game of this auxiliary game in the sense of Forges (1986) (see also Myerson (1986)). Thus, her revelation principle applies.

[13]Here, we focus on a deterministic recommendation of $\sigma$, rather than any stochastic recommendation. Accordingly, $\delta$ is denoted simply by $\delta(s)$ instead of $\delta(r, s)$. Our results do not change with stochastic recommendations.

That is, each $i$ must find it optimal to obey the recommended $\sigma_i$ and report the realized $s_i$ truthfully.

Although the constraints are concisely summarized by the simple-looking inequalities above, they can potentially be very complicated. First, changing $\sigma_i$ affects the joint distribution $\alpha$ of $(\theta, s)$ and the agent's cost, but the way it affects $\alpha$ is somewhat limited because $i$ cannot affect $-i$'s information structure. In particular, the marginal of $\alpha$ on $(\theta, s_{-i})$ must not change at all. Second, an agent may potentially want to make a double deviation, that is, changing $\sigma_i$ and at the same time changing his reporting strategy.

**Remark 1.** *Here, we do not explicitly impose each agent's individual rationality/participation constraint. It is not difficult to accommodate this constraint: let us require that any feasible direct mechanism must have an extra message $\phi_i$ (a "non-participation" message) so that $i$'s message space is now $S_i \cup \{\phi_i\}$, and $\delta(\phi_i, m_{-i})$ is some specific allocation (a "non-participation allocation") for agent $i$, for any given $m_{-i}$. Accordingly, the individual rationality constraints, both at the ex ante and ex interim stages are captured as a part of the above incentive compatibility constraints.*[14]

# 3 Main result

Fix any function $d^* : \Theta \to D$, which describes all the economically relevant outcomes in this environment except for the information acquisition costs. If the principal could observe $\theta$, then any $d^*$ is attainable without any information acquisition cost on the agents' side. In this sense, one may interpret this $d^*$ together with zero cost as the *first-best* outcome.[15]

In this section, for any given $d^*$, we explicitly construct a mechanism that implements $d^*$ with zero cost. That is, the first-best outcome is indeed attainable, *even though the principal cannot directly observe $\theta$.*

[14] Ex interim individual rationality is guaranteed because agent $i$ can always deviate to $\tau_i(\cdot) \equiv \phi_i$. Ex ante individual rationality is guaranteed because agent $i$ can always deviate to a costless $\sigma_i$ and then to $\tau_i(\cdot) \equiv \phi_i$.

[15] For example, one may imagine the principal's own preference and define $d^*(\theta)$ as his best decision given $\theta$.

**Theorem 1.** *Fix any $d^* : \Theta \to D$. Under Assumption 1, there exists a mechanism $(\sigma, \delta)$ such that (i) $\sum_s \alpha(\theta, s) 1_{\{\delta(s)=d^*(\theta)\}} = \mu_0(\theta)$ for all $\theta$, and (ii) $c_i(\sigma_i) = 0$ for all $i$.*

*Proof.* Our construction directly builds on Zhu (2021):

**Lemma 1.** *(Zhu (2021); Lemma 2)*[16] *Fix any $M, N \in \mathbb{N}$ and any random variable with finite support, $\tau_0 \in \{0, \ldots, K-1\}$. There exists $\tau_1, \ldots, \tau_N$ such that (i) for any $i \in \{1, \ldots, N\}$, $(\tau_0, \tau_i)$ are pairwise independent, and (ii) for any $i, j, k \in \{1, \ldots, N\}$ with $i \neq j$, $\tau_0 | (\tau_i, \tau_j)$ and $\tau_k | (\tau_i, \tau_j)$ are degenerate.*

In our context, let $\tau_0 = \theta$ and $s_i = \tau_i$, and let $\sigma_i \in \Sigma_i$ be the one that induces $s_i$. Then, the first property says that $\theta, s_i$ are pairwise independent, implying $c_i(\sigma_i) = 0$. The second property says that, given $s_i, s_j$ with $i \neq j$, we can fully identify $\theta$ and any $s_k$ without error, that is, there exist $\theta(s_i, s_j)$ and $s_k(s_i, s_j)$ such that:

$$\Pr(\theta = \theta(s_i, s_j) | s_i, s_j) = \Pr(s_k = s_k(s_i, s_j) | s_i, s_j) = 1.$$

See Zhu (2021) for the detailed construction and proof.

Let the principal recommend the above $\sigma$, and offer the decision rule $\delta$ as follows: $\delta(s) = d^*(\theta)$ if (i) for any $i, j$ with $i \neq j$, we have

$$\theta = \theta(s_i, s_j);$$

or if (ii) there is $i$ such that, for any $j, k$ where $i, j, k$ are all different, we have

$$\theta = \theta(s_j, s_k).$$

In any other case, $\delta(s)$ is arbitrary.

Clearly, if the agents obey the recommendation and report their signals truthfully, then the first-best outcome is attained. Therefore, we complete the proof by showing that it satisfies incentive compatibility. Take any agent $i$, and suppose that he deviates to any $\sigma_i'$ and report $\tau_i(s_i)$ if each $s_i$ is realized. First, his cost of information acquisition increases weakly. Second, his reporting decision does not affect the social decision at all, because the principal executes

[16] Zhu (2021) calls the combination of these properties *individually uninformative but aggregately revealing.*

$\delta(s) = d^*(\theta(s_j, s_k))$ for an arbitrary pair $(j, k)$ which does not include $i$. Therefore, it is incentive compatible. $\square$

# 4 Application: Full-surplus extraction in common value auctions

Consider the following common value auction environment. The seller (principal) has a single indivisible good, and there are $I \geq 4$ bidders. The value of the good is common to all the bidders, denoted by $\theta \in \Theta$, where $\Theta$ is finite. In fact, the analysis of this section can be straightforwardly extended to the case of "non-pure" common values where each $i$'s valuation is $v_i(\theta)$. Let $\mu_0(\theta)$ denote the probability that $\theta$ is the bidders' common value.

Each bidder $i$'s payoff is $\theta q_i - t_i - c_i(\sigma_i)$ if he wins the good with probability $q_i$, pays $t_i$ to the seller, and spends $c_i(\sigma_i)$ as his information acquisition cost. In case he does not participate in the mechanism, his outside-option payoff is 0. The seller's payoff is revenue, $\sum_{i=1}^{I} t_i$.

The first-best expected surplus of this society is the expected common value:

$$\sum_{\theta \in \Theta} \mu_0(\theta)\theta = \mathbb{E}[\theta].$$

There are several cases where the seller can easily earn $\mathbb{E}[\theta]$. First, if the seller *knows* $\theta$, then he can simply post price $\theta$. Even if the seller does not know $\theta$, if the bidders know $\theta$ as their common knowledge (i.e., as *free* information), then again the seller can earn $\mathbb{E}[\theta]$. Conversely, if all the bidders are completely *uninformed* (so that each only knows the common prior $\mu_0$), then again, the seller can post price $\mathbb{E}[\theta]$.

Notice that, with costly information acquisition as considered in our paper, neither of the above ideas would work. First, although it might be possible to make every bidder fully learn $\theta$ in some equilibrium, it does not yield $\mathbb{E}[\theta]$ as long as full information is strictly costly. Second, if the seller posts price $\mathbb{E}[\theta]$, then each bidder has a strong incentive of knowing whether the true $\theta$ is below $\mathbb{E}[\theta]$ or not: If $i$ finds that $\mathbb{E}[\theta|s_i] < \mathbb{E}[\theta]$ given some signal $s_i$, he would not buy the good. As long as such information is not too costly, the bidder would be better off by acquiring it.

Therefore, with a general information acquisition cost function, the equilibrium information should be somewhere between full and no information, and it is *a priori* unclear how the seller should find the optimal balance of information and rent extraction. Nevertheless, as long as the cost functions satisfy Assumption 1, Theorem 1 implies that the full-surplus extraction is possible.

**Corollary 1.** *Under Assumption 1, there is a mechanism which yields* $\mathbb{E}[\theta]$ *as the seller's expected revenue (and each bidder earns 0).*

It is worth emphasizing that the logic here is very different from that of Crémer and McLean (1988). In their paper, the seller exploits an *exogeneously given* correlated signal structure, in order to construct a side-bet scheme that extracts the entire surplus. In our case, each bidder can choose any information structure. Indeed, if he prefers, a bidder can choose an information structure such that his information is *independent* from all the other bidders' signals. The Crémer-McLean lottery scheme, therefore, does not work here. Also, in their mechanism, each bidder's payoff can be strictly negative *ex post*, while in our case, it is zero *ex post*. Indeed, if a mechanism offers a negative ex post payoff in our environment, a bidder would have a strong incentive of getting a signal indicative of that event and then report differently, in order to avoid such a negative payoff.

# 5 Concluding remarks

It is quite natural that agents may desire to refine their information in response to the mechanism. This paper proposes one possible framework, based on "private-value" cost functions. We show that such a specification leads to an extremely positive result. We conclude the paper with two remarks.

## 5.1 Interdependent cost functions

One natural criticism may be that our mechanism induces full information if the signals are aggregated, even though any single signal is completely uninformative: would it be reasonable

to assume that such $\sigma_i$ is costless? Because the answer is necessarily yes under Assumption 1, the question is essentially whether Assumption 1 itself is reasonable.

As far as we are aware of, there is no clear consensus nor guidance as to what kinds of correlation structures of the signals are more or less expensive to each agent. There seem to be cases where more positive correlation is more expensive. For example, fix agent 1's private information, and consider agent 2. If acquiring a positively correlated information necessarily means that agent 2 must *steal* (perhaps a part of) agent 1's information, more positive correlation would be more costly. there are however opposite situations. Strulovici (2021), for example, considers the environment where hard evidence is scarce in the sense that, if one agent "picks up" a piece of evidence, then it becomes difficult for the others to get the same or similar evidence. There also seem to be cases where *less correlated* signals are more costly: for example, imagine that there exist 2 newspapers, and $\sigma_i$ corresponds to the decision of which newspapers to buy (including the options of buying both or none). Assume that the information in those newspapers is highly positively correlated, and assume that agent 1 buys (only) newspaper 1 for sure. In this case, agent 2 receives a positively correlated signal regardless of the newspaper he chooses, while in order to obtain something less correlated, he must buy *both* newspapers and orthogonalize newspaper 2's information from newspaper 1's information. Thus, less correlation is more costly.

This discussion suggests that we must be more specific about the microstructure of the information sources, in order to determine which correlation structures are more costly. Mechanism design with such more specific information acquisition cost structures would certainly be an interesting future direction, and we hope this article could serve as a first step toward that direction.

## 5.2 Two or three agents

Obviously, the individually-uninformative-but-aggregately-revealing signal structures require four or more agents. With three agents, although it is possible to determine whether *some* agent has unilaterally deviated or not, but it is not possible to identify who is the deviator (and hence

to identify the true $\theta$). If there exists $d \in D$ that can serve as a severe punishment for all agents for any given $\theta$, then the same result obtains. See Zhu (2021) too.

With two agents, then each agent has much more freedom. Indeed, the authors work on a separate project with two agents. There, even under Assumption 1, an extremely positive result similar to the one obtained in this paper does not generally hold. The optimal mechanism typically involves some costly information acquisition, and hence, the specification of the cost functions matters even more.

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